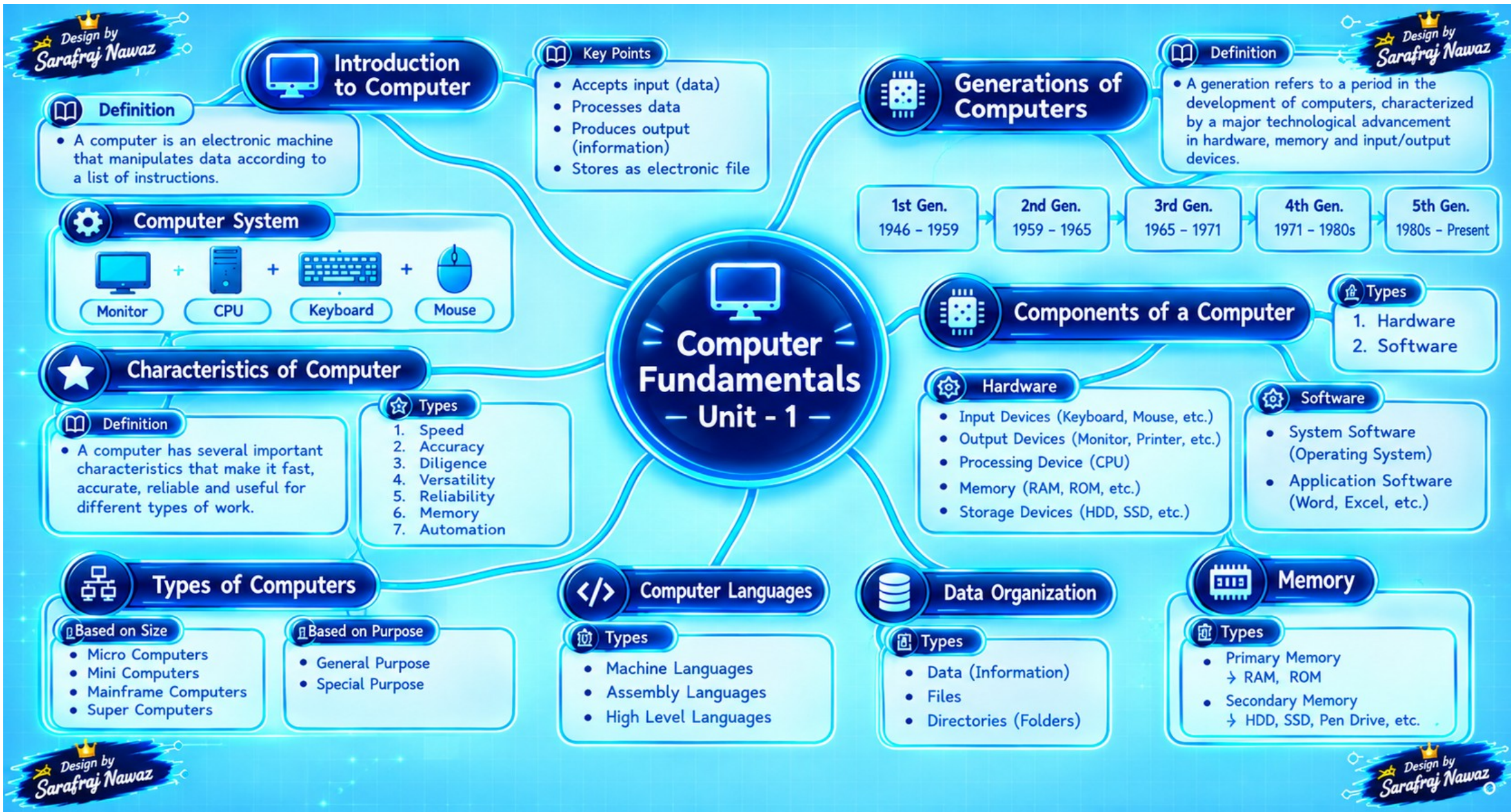


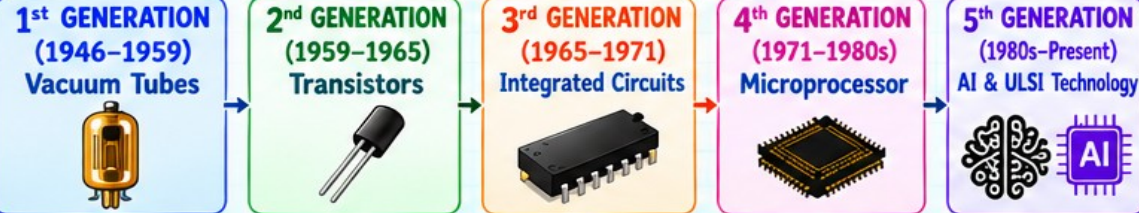


GENERATION OF COMPUTERS

Page 1/2

Computers have been developed through different stages called generations computers. Each generation is based on the major technology used to build computers.

EVOLUTION TIMELINE



1 FIRST GENERATION (1946–1959)



Examples:
ENIAC, UNIVAC I,
IBM 701, Mark I

Definition

- The first generation of computers used vacuum tubes as the main electronic component. These computers were very large, consumed a lot of electricity, and generated much heat.

Main Features

- Used vacuum tube technology.
- Magnetic drums were used for memory.
- Very large in size.
- Required large amount of electricity.
- Used punch cards for input.

Advantages

- Very fast compared to previous calculating devices.
- Could perform calculations in milliseconds.
- Made large scale electronic computing possible.

Disadvantages

- Very large in size.
- Consumed a lot of electricity.
- Generated excessive heat.
- Required air conditioning.
- Frequent maintenance.
- Vacuum tubes failed often.
- Very expensive.
- Not portable.
- Supported only machine language.
- Magnetic drums had limited storage.

2 SECOND GENERATION (1959–1965)



Examples:
IBM 1400 series,
IBM 1600 series,
UNIVAC III

Definition

- The second generation of computers used transistors instead of vacuum tubes. They were smaller, faster, more reliable and consumed less power.

Main Features

- Used transistors.
- Smaller, faster and more reliable.
- Magnetic core memory used.
- Required less heat.
- Faster than first generation.
- More affordable.
- Supported high level languages (FORTRAN, COBOL, etc.).
- Punch cards used for input.
- Batch processing was used.

Advantages

- Smaller than first generation.
- More reliable.
- Consumed less electricity.
- Generated less heat.
- Faster than first generation.
- Easier to maintain.
- Supported high level languages.

Disadvantages

- Required cooling systems.
- Regular maintenance required.
- Manufacturing was complex.
- Used for specific applications.
- Costly.
- Punch cards still used for input.

3 THIRD GENERATION (1965–1971)



Examples:
IBM System/360,
IBM System/370,
UNIVAC 1108

Definition

- The third generation of computers used integrated circuits (ICs). ICs combined many components on a single chip, making computers smaller, faster and more reliable.

Main Features

- Used Integrated Circuits (ICs).
- Computers became smaller and more reliable.
- Semiconductor memory used.
- Magnetic disks and floppy disks used for storage.
- Keyboard used for input and monitor for output.
- Advanced operating systems.

Advantages

- Smaller than second generation.
- More reliable.
- Faster processing.
- Consumed less electricity.
- Generated less heat.
- Less expensive.
- Suitable for general purpose applications.

Disadvantages

- Manufacturing ICs required sophisticated technology.
- IC based systems were still costly.
- Maintenance required skilled technicians.

4 FOURTH GENERATION (1971–1980s)



Examples:
IBM PC, Osborne 1,
Apple II, Star 1000

Definition

- The fourth generation of computers used (large-scale integrated circuits) technology. These computers were more compact in this generation.

Main Features

- Used LSI and VLSI technology.
- Personal computers became popular.
- Reside onto hard disk usage.
- GUI (Graphical User Interface).
- Keyboard and mouse used.
- Monitor used for output.
- Multitasking and development of operating systems.

Advantages

- Very small in size.
- Faster processing.
- More reliable.
- Consumes less electricity.
- Generated very little heat.
- Portable computers available.
- Personal computer widely available.

Disadvantages

- Advanced technology required for manufacturing.
- Skilled professionals required.
- Rapid changes made older systems obsolete.

EVOLUTION OF COMPUTERS

(GENERATION WISE)

Page 2/2

Generation Wise Comparison.

Generation	Years	Main Technology	Size	Speed	Memory	Examples
1st	1946–1959	Vacuum Tubes	Very Large	Very Slow	Very Small	ENIAC, UNIVAC I
2nd	1959–1965	Transistors	Large	Faster	Small	IBM 1401, UNIVAC II
3rd	1965–1971	Integrated Circuits	Smaller	Fast	Medium	IBM System/360
4th	1971–1980s	Microprocessor (LSI/VLSI)	Small	Very Fast	Large	IBM PC, Apple II
5th	1980s–Present	AI & ULSI	Very Small	Extremely Fast	Very Large	Laptops, Smartphones



From Vacuum Tubes to AI – The Journey of Computers
Keeps Getting Smarter!

1. WHAT IS A COMPUTER?

BCA | Unit-I | Introduction to Computer

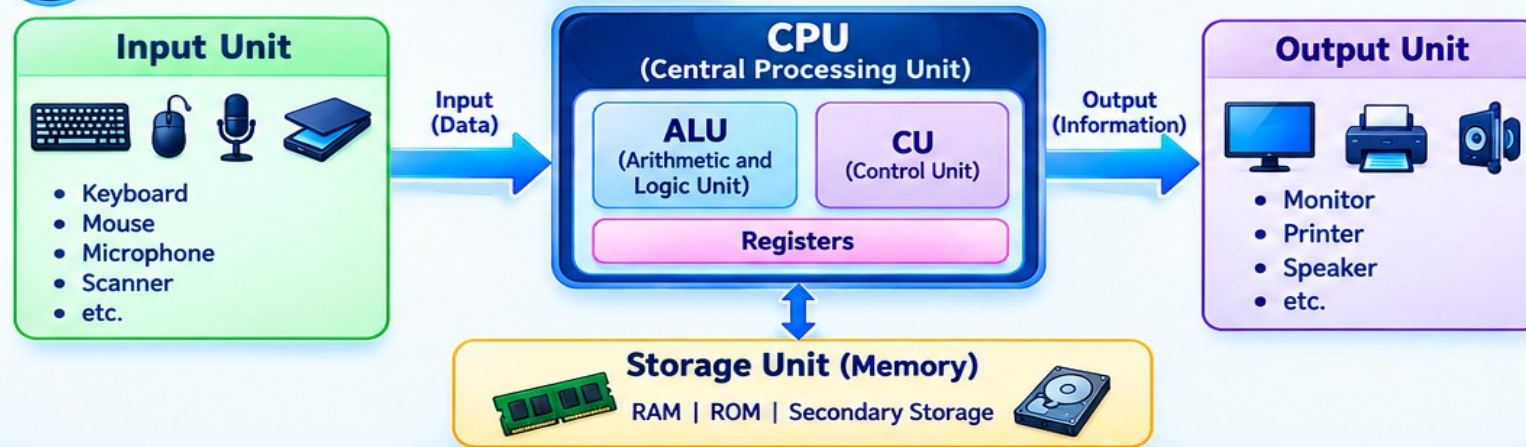


Definition

A **computer** is an electronic machine that accepts data as input, processes it according to a set of instructions, produces meaningful information as output, and stores the results for future use.



Block Diagram of a Computer



Key Points

- 1 Accepts data as input
- 2 Processes data using instructions
- 3 Produces meaningful information as output
- 4 Stores the results for future use



Examples of Input, Output and Storage Devices



Uses of a Computer

- Education
- Business
- Communication
- Banking
- Research
- Entertainment
- and many more...



CHARACTERISTICS OF COMPUTER

A computer has several important characteristics that make it fast, accurate, reliable and useful for different types of work.

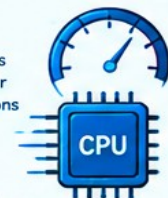
1 Speed

Definition:

A computer can perform calculations and process instructions much faster than a human - millions or even billions of operations in a very short time.

Examples:

- Calculating large numbers
- Processing big data
- Running complex programs



Millions of operations per second

2 Accuracy

Definition:

A computer produces highly accurate results when the input and instructions are correct. Most errors occur due to wrong input or incorrect instructions.

Examples:

- Financial calculations
- Scientific research
- Data processing



100% Accurate (when input is correct)

3 Diligence

Definition:

A computer can perform the same task repeatedly without getting tired, bored or losing concentration - maintaining the same speed and accuracy throughout.

Examples:

- Repetitive data entry
- Continuous monitoring
- 24/7 operations



No fatigue
No loss of concentration

4 Versatility

Definition:

A computer can perform many different tasks - preparing documents, watching videos, browsing the internet, programming, designing and managing databases.

Examples:

- Word processing
- Web browsing
- Graphic design



Many tasks in one device

5 Reliability

Definition:

A computer provides consistent and dependable results - given the same input and instructions under the same conditions, it will normally produce the same output every time.

Examples:

- Accurate data storage
- Stable performance
- Minimal errors



Consistent Results

6 Memory

Definition:

A computer can store a large amount of data, instructions and results and retrieve them whenever required. Computer memory is mainly divided into primary and secondary memory.

Examples:

- RAM (temporary)
- Hard Disk (permanent)
- Quick data access



Stores - Remembers - Retrieves

CHARACTERISTICS OF COMPUTER (contd.)

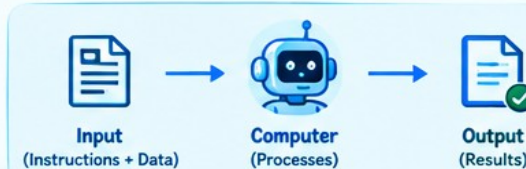
7 Automation

Definition:

Once a computer is given the required instructions and data, it can perform tasks automatically with little or no human intervention.

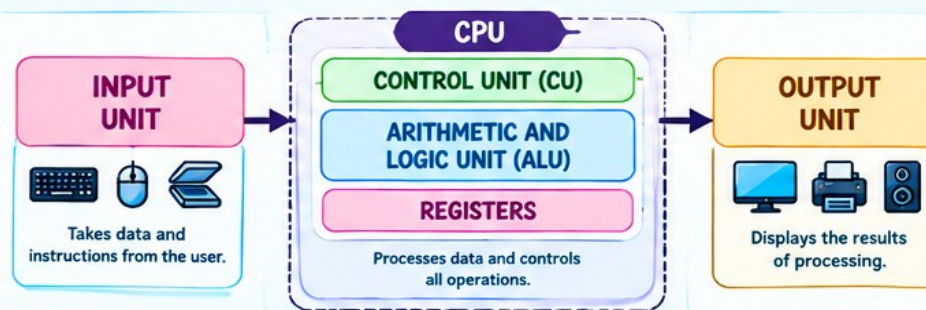
Examples:

- Automatic calculations
- Scheduled tasks
- Industrial automation



BLOCK DIAGRAM OF COMPUTER

The block diagram of a computer shows the main functional units and how they are interconnected to perform various operations.

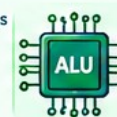


CPU - ITS COMPONENTS

1 ALU (Arithmetic and Logic Unit)

Definition:

The ALU is the part of the CPU that performs arithmetic operations (such as addition, subtraction, multiplication, division) and logical operations (such as AND, OR, NOT, XOR, etc.) on data.



2 CU (Control Unit)

Definition:

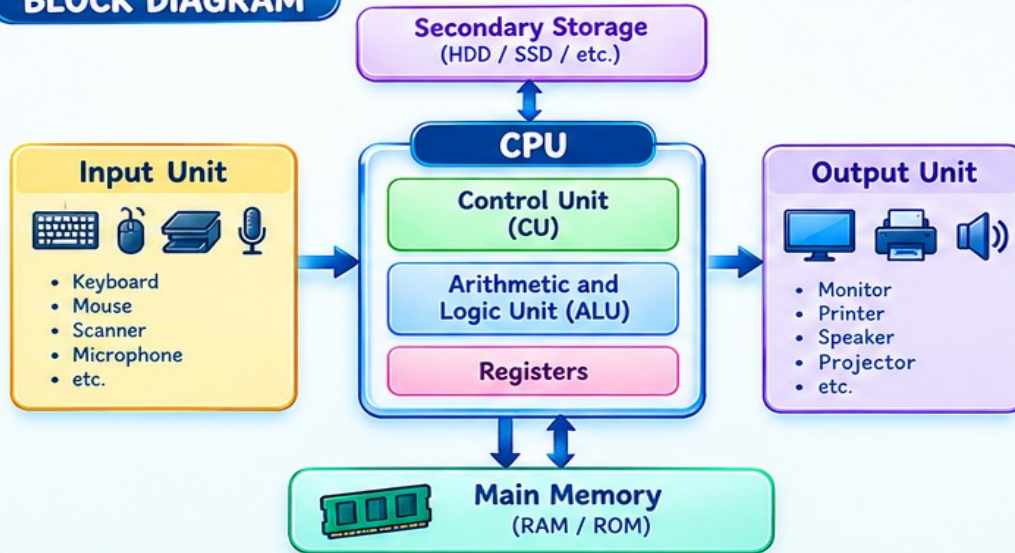
The CU controls and coordinates all the activities of the computer system. It fetches instructions from memory, decodes them and sends control signals to other units (ALU, Memory, Input/Output devices) to execute the instructions.



BLOCK DIAGRAM OF COMPUTER

The block diagram of a computer shows the main functional units and how they are interconnected to perform various operations.

BLOCK DIAGRAM



FUNCTIONS OF EACH UNIT

Unit	Function	Example Devices
Input Unit	Accepts data and instructions from the user and converts them into machine-readable form.	Keyboard, Mouse, Scanner, Microphone
CPU	Processes data, performs calculations and controls all operations.	ALU, CU, Registers
Memory	Stores data and instructions for current and future use.	RAM, ROM, Cache
Output Unit	Displays the processed results in a human-readable form.	Monitor, Printer, Speaker
Secondary Storage	Stores data permanently even when power is off.	Hard Disk, SSD, Pen Drive

CPU – ITS COMPONENTS

The CPU (Central Processing Unit) is the brain of the computer system. It consists of three main parts: ALU, CU and Memory (Registers).

1 ALU (Arithmetic and Logic Unit)

Definition:

The ALU is the part of the CPU that performs arithmetic operations (such as addition, subtraction, multiplication, division) and logical operations (such as AND, OR, NOT, XOR, etc.) on data.

It performs:

- Arithmetic operations (+, -, ×, ÷)
- Logical operations (AND, OR, NOT, XOR)
- Comparison operations (>, <, =, etc.)



2 CU (Control Unit)

Definition:

The CU controls and coordinates all the activities of the computer system. It fetches instructions from memory, decodes them and sends control signals to other units (ALU, memory, input/output devices) to execute the instructions.

It performs:

- Fetches instructions from memory
- Decodes the instructions
- Generates control signals
- Coordinates with ALU, Memory and I/O devices



3 MEMORY (Registers)

Definition:

Registers are small, high-speed storage locations inside the CPU. They hold data, instructions and intermediate results while processing.

Types:

- Accumulator (stores results)
- Program Counter (holds next instruction address)
- Instruction Register (stores current instruction)
- General Purpose Registers (store temporary data)



KEY FEATURES OF CPU



Fast Processing



High Accuracy



Handles Multiple Tasks



Small Size



Low Power



TYPES OF COMPUTERS AND THEIR FEATURES



Computers can be classified into different types based on their size, processing speed, storage capacity, performance and number of users.

1 MINI COMPUTERS

Definition :

A Mini Computer is a medium-sized computer that is more powerful than a microcomputer but smaller and less powerful than a mainframe computer.

Features :

- Medium processing speed.
- Supports multiple users at the same time.
- More powerful than microcomputers.
- Has greater storage capacity than microcomputers.
- Used in small and medium-sized organizations.
- Can perform multiple tasks simultaneously.

Examples : PDP-11, IBM AS/400.



Exam Point : Mini Computer → Medium-sized, multi-user and more powerful than a microcomputer.



2 MICRO COMPUTERS

Definition :

A Micro Computer is a small computer that uses a microprocessor as its CPU and is generally designed for use by one person at a time.

Features :

- Small in size.
- Low cost compared with larger computers.
- Designed mainly for a single user.
- Easy to operate.
- Requires less space.
- Suitable for personal and office use.

Examples : Desktop Computer, Laptop, Tablet, Smartphone.



Exam Point : Micro Computer → Small, low-cost and generally single-user computer.



3 MAINFRAME COMPUTERS

Definition :

A Mainframe Computer is a large and powerful computer designed to process huge amounts of data and support many users simultaneously.

Features :

- Very high processing capacity.
- Supports hundreds or thousands of users.
- Large storage capacity.
- Highly reliable and secure.
- Can process large volumes of data.
- Used by banks, government organizations, airlines, and large companies.

Examples : IBM zSeries mainframes.



Exam Point : Mainframe Computer → Powerful multi-user computer used for large-scale data processing.



4 SUPER COMPUTERS

Definition :

A Super Computer is the fastest and most powerful type of computer, designed to perform extremely complex calculations and process huge amounts of data at very high speed.

Features :

- Extremely high processing speed.
- Very large storage capacity.
- Performs complex calculations very quickly.
- Uses parallel processing.
- Very expensive to build and maintain.
- Used for scientific and engineering applications.

Applications :



Weather forecasting



Space research



Scientific research



Nuclear research



Climate modelling



AI and Simulations



Exam Point : Super Computer → Extremely powerful and fastest computer used for complex calculations.



COMPARISON TABLE

Type	Size	Users	Processing Speed	Storage Capacity	Main Feature	Common Use
Micro Computer	Small	Usually one	Low to Medium	Small	Personal use	Home, Office
Mini Computer	Medium	Multiple	Medium	Medium	Medium-scale processing	Organizations
Mainframe	Large	Many	High	Large	Huge data processing	Banks, Airlines, Large Companies
Super Computer	Very Large / High-performance	Specialized	Very High	Very Large	Extremely fast calculations	Research, Weather, Scientific Work

Remember

- Micro → Personal
- Mini → Medium / Multiple Users
- Mainframe → Massive Data / Many Users
- Super → Scientific / Highest Speed

ADVANTAGES AND DISADVANTAGES OF COMPUTER



Definition

The advantages of computers are the benefits provided by computers, while the disadvantages are the limitations or negative effects of using computers.



ADVANTAGES OF COMPUTER

- 1 **High Speed**
Computer can perform millions of calculations in a very short time.
- 2 **Accuracy**
It gives highly accurate results when correct data and instructions are provided.
- 3 **Large Storage**
It can store a huge amount of data and information.
- 4 **Automation**
Computers can perform repetitive tasks automatically.
- 5 **Multitasking**
It can perform multiple tasks at the same time.
- 6 **Reliability**
It can work continuously without getting tired.
- 7 **Easy Communication**
It helps us communicate through email, video calls, and messaging.
- 8 **Time Saving**
It completes complex tasks much faster than humans.
- 9 **Easy Access to Information**
It provides quick access to information through the internet.
- 10 **Useful in Education**
It helps students with online classes, notes, research, and presentations.
- 11 **Useful in Business**
It helps in accounting, billing, management, and data analysis.
- 12 **Entertainment**
It can be used for movies, music, games, and other entertainment.



"Smarter Computers for a Better Future"



Key Takeaway

- ✓ Computers make our work faster, easier and more efficient.
- ✓ They are useful in almost every field of life.
- ✓ But we should also be aware of their limitations and risks.



In Short (Exam Point)

Advantages = Benefits
Disadvantages = Limitations

ADVANTAGES AND DISADVANTAGES OF COMPUTER



DISADVANTAGES OF COMPUTER

- 1 **Health Problems**
Excessive use can cause eye strain, back pain, and other health issues.
- 2 **Cybercrime**
Computers can be used for hacking, fraud, and other cybercrimes.
- 3 **Security Risks**
Viruses and malware can damage data and systems.
- 4 **Privacy Problems**
Personal information can be stolen or misused.
- 5 **Unemployment**
Automation can reduce the need for human workers in some jobs.
- 6 **Dependency**
Excessive dependence on computers can reduce human skills and productivity.
- 7 **Costly**
Good computers and their maintenance can be expensive.
- 8 **Power Consumption**
Computers require electricity to operate.
- 9 **Data Loss**
Data may be lost due to hardware failure, viruses, or accidental deletion.
- 10 **Time Wastage**
Excessive use of games, social media, and entertainment can waste time.
- 11 **Lack of Emotions**
Computers cannot think or feel emotions like humans.
- 12 **Technical Problems**
Hardware or software failures can interrupt work.



Important Points

- Computers are powerful tools but not perfect.
- They are useful when used wisely and safely.
- Overuse can harm our health, privacy and security.
- Balance is the key to getting the maximum benefit from computers.



Quick Revision

Advantages (8)

Speed, Accuracy, Storage, Automation, Multitasking, Reliability, Communication, Time Saving.

Disadvantages (8)

Health Problems, Cybercrime, Security Risks, Privacy Issues, Unemployment, Dependency, Cost, Data Loss.



"Use Computer for Progress, Not for Addiction"

★ (Machine Languages, Assembly Languages, High Level Languages)

LANGUAGE

- Any medium to communicate is called a language.

PROGRAMMING LANGUAGE

- Any language that is used to communicate with a computer or a machine is called Programming language. Programming Language is in the form of a set of instructions.

TYPES OF PROGRAMMING LANGUAGE

- Low Level / Binary language [0,1]
- Assembly / Mid / Bit code language
- High Level language



1. LOW LEVEL LANGUAGE

- Low level Programming language is the first programming language, which was used to instruct computers. It is in binary form [0,1], so it is called Binary language. A computer can directly understand low level language but it's difficult for a human to understand.

Example :

Code

```
10110010 11001100 10101010
11001010 10110011 11001101
10101010 11001010 10110011
11001100 10101010 11001010
```



Computer can understand directly

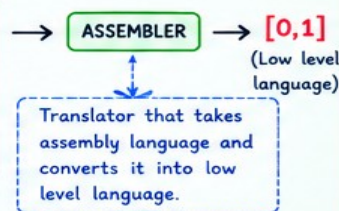
2. ASSEMBLY / MID / BIT CODE LANGUAGE

- Mid level language is called Assembly level language. In Assembly level language, predefined words are used to make programming easy. In Assembly level programs, work was easier than low level language, but computers can't understand assembly level language directly, and also not humans.
- A computer can't understand assembly level language directly, so an Assembler is used in between as a translator, which takes assembly language and converts it into low level language.

Example :

Code

```
MOV A, B
ADD A, C
SUB A, D
MUL A, E
DIV A, F
```



COMPARISON TABLE

Type	Form	Understand by	Translator Used	Example
Low Level / Binary	Binary [0,1]	Computer	No Translator (Directly)	Machine Code
Assembly / Mid / Bit Code	Predefined words (Mnemonics)	Neither Computer nor Human Easily	Assembler	MOV, ADD, SUB
High Level	Basic English (Easy Words)	Human	Compiler / Interpreter	C, Java, Python

★ (Machine Languages, Assembly Languages, High Level Languages)

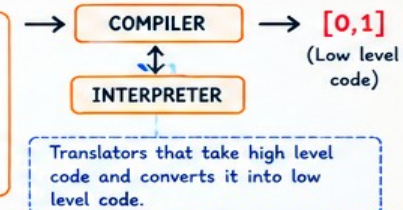
3. HIGH LEVEL LANGUAGE

- A language that a human can understand directly is called high level language. In high level language basic English is used, which makes programming very easy to understand.

Example :

Code

```
int a = 10;
int b = 20;
int sum = a + b;
print(sum);
```



2. KEY POINTS

- A language is a medium to communicate.
- Programming language is used to communicate with a computer or machine.
- Low level language is binary [0,1].
- Assembly language uses predefined words.
- High level language uses basic English.

1. TYPES OF LANGUAGE

- Low Level / Binary language [0,1]
- Assembly / Mid / Bit code language
- High Level language

WHY USE HIGH LEVEL LANGUAGE?

- Easy to understand
- Less coding
- More readable
- Portable

EXAMPLES OF HIGH LEVEL LANGUAGES

- C
- C++
- Java
- Python
- JavaScript



KEY DIFFERENCE (IN SHORT)

Feature	Low Level	Assembly	High Level
Language Type	Binary (0,1)	Mnemonics	English-like
Ease of Use	Difficult	Moderate	Easy
Human Readable	No	Partially	Yes
Directly Understood by Computer	Yes	No (Needs Assembler)	No (Needs Compiler/Interpreter)
Example	101010	MOV A, B	C, Java, Python

DATA ORGANIZATION

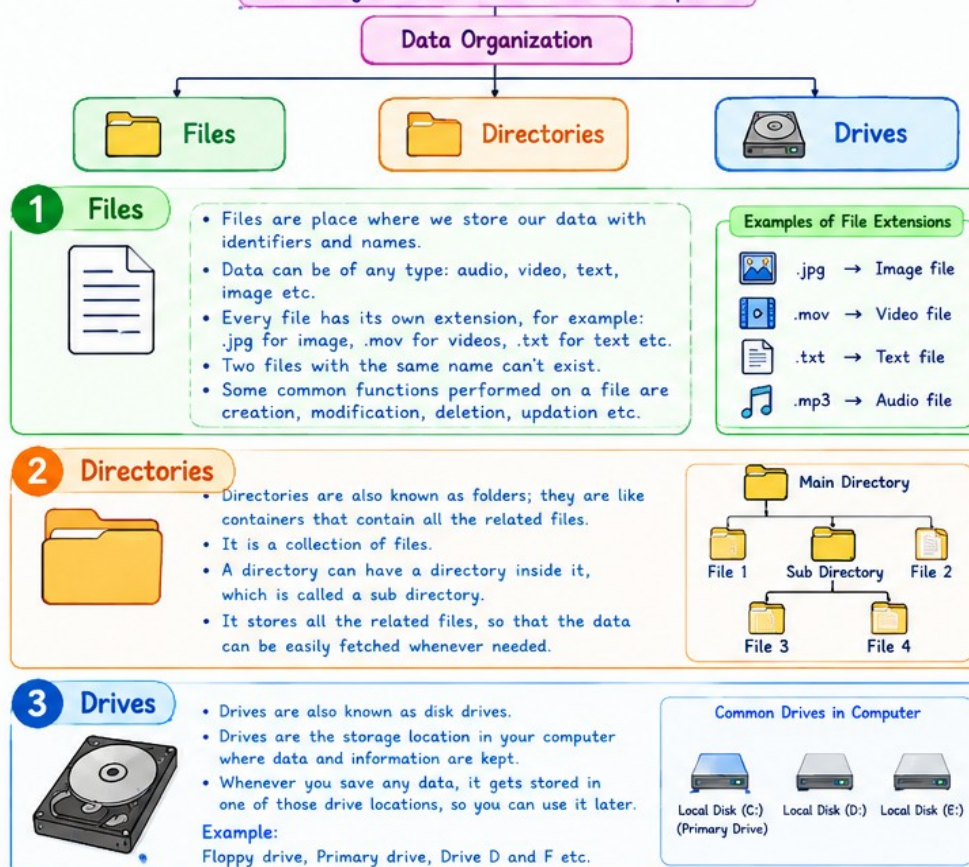
or Data Organization, Drives, Files, Directories



What is Data Organization?

Data organization is the way of arranging data in the most efficient manner so that it can be easily accessed whenever required. It is the arrangement of data in the most possible orderly and logical manner. Data can be arranged into sheets with headings or into graphs, charts, or tables.

Data Organization is divided into three parts:



Types of Files

Type	Description
Text File	Contains plain text data.
Image File	Contains pictures or graphics.
Audio File	Contains sound data.
Video File	Contains video data.
Executable File	Contains program instructions.
Compressed File	Contains compressed data (e.g., .zip, .rar).

Types of Directories (Folders)

Type	Description
Root Directory	The top most directory in a file system.
Sub Directory	A directory inside another directory.
Parent Directory	The directory above the current directory.
Child Directory	A directory inside a directory.
Home Directory	The default directory for a user.

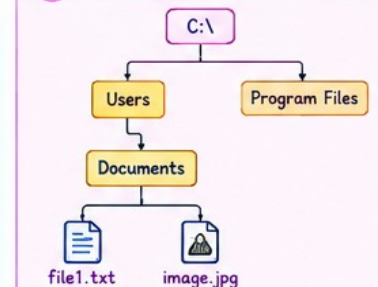
Types of Drives

Type	Description
Hard Disk Drive (HDD)	Large storage capacity, non-volatile, magnetic storage.
Solid State Drive (SSD)	Faster than HDD, no moving parts.
Optical Drive	Reads/writes CDs and DVDs.
Removable Drive	Portable storage devices (USB, Pen Drive, Memory Card).

File Naming Conventions

- A file name consists of two parts: name and extension.
- The extension tells us the type of file.
- Example: photo.jpg (name: photo, extension: .jpg)
- We should use meaningful names and avoid special characters like / \ : * ? " < > |

Directory Structure Example



Key Points

- Files store data, directories organize files, and drives store data permanently.
- Files have extensions that define their type.
- Directories can contain subdirectories.
- Drives are the physical storage devices of the computer.
- Proper organization helps in easy access, management and security of data.

Well organized data saves time and effort!



TYPES OF MEMORY

(Primary And Secondary)

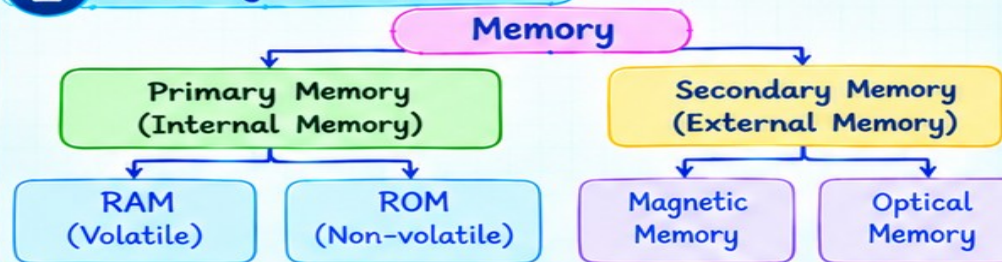
RAM, ROM, PROM, EPROM, EEPROM, I/O Devices
(Scanners, Plotters, LCD, Plasma Display)

1 What is Memory?

- Memory is a storage device which is used to store data for further use.
- It can be broadly classified into Primary and Secondary memory.
- Memory helps the computer to store data, instructions and programs.



2 Memory Classification



3 Primary Memory

- Primary memory is directly accessible by the CPU.
- It is faster than secondary memory.
- It is of limited capacity.
- It is used to store data and instructions temporarily.

Types of Primary Memory

RAM (Random Access Memory)

- Volatile memory (data is lost when power is turned off).
- Used for temporary storage.
- Fast access speed.

Types of RAM:

1. **SRAM** (Static RAM)
2. **DRAM** (Dynamic RAM)



ROM (Read Only Memory)

- Non-volatile memory (data is retained even when power is off).
- Stores permanent instructions for booting the system.
- Slower than RAM.

Types of ROM:

1. **PROM** (Programmable ROM)
2. **EPROM** (Erasable Programmable ROM)
3. **EEPROM** (Electrically Erasable Programmable ROM)



4 ROM Types

1. PROM (Programmable ROM)

- Can be programmable only once by the user.
- After programming, data cannot be changed or erased.
- Example: PROM chips.



2. EPROM (Erasable Programmable ROM)

- Can be erased and reprogrammed using ultraviolet (UV) light.
- It can be programmed multiple times.
- Example: EPROM with UV window.



3. EEPROM (Electrically Erasable Programmable ROM)

- Can be erased and reprogrammed electrically.
- Faster than EPROM.
- Used in modern devices (BIOS, flash memory).
- Example: EEPROM, Flash memory.



ROM is non-volatile memory, so the data is not lost when power is turned off. It is slower than RAM but stores permanent data.

5 Secondary Memory

- Secondary memory is used to store large amount of data and information permanently.
- It is non-volatile memory (data is not lost when power is off).
- It has higher storage capacity than primary memory.
- It is cheaper than primary memory.

Types of Secondary Memory

Magnetic Memory

- Stores data using magnetic patterns.
- Examples: Hard Disk, Floppy Disk, Magnetic Tape.



Hard Disk



Floppy Disk



Magnetic Tape

Optical Memory

- Stores data using laser or light technology.
- Examples: CD, DVD, Blu-ray Disc.



CD



DVD



Blu-ray

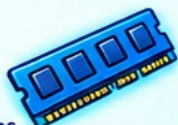


Primary memory is the main memory of a computer. It stores data and instructions that are currently being processed by the CPU. The CPU can access it directly.

Types of Primary Memory

RAM (Random Access Memory)

- Volatile memory.
- Stores data temporarily.
- Fast speed.
- Used for running programs and processing data.



Examples : DDR3, DDR4, DDR5

ROM (Read Only Memory)

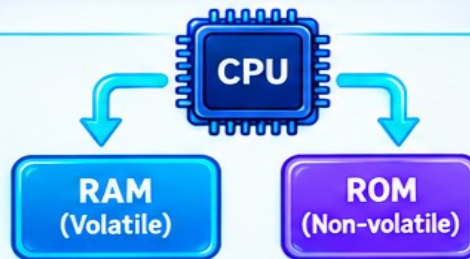
- Non-volatile memory.
- Stores permanent data (firmware).
- Slower than RAM.
- Used for booting the system and basic instructions.



Examples : PROM, EPROM, EEPROM

Features of Primary Memory

- ✓ Directly accessible by CPU.
- ✓ Faster than secondary memory.
- ✓ Smaller capacity.
- ✓ More expensive per GB.
- ✓ Includes both volatile and non-volatile memory.



Advantages of Primary Memory

- ✓ Provides quick access to data and instructions.
- ✓ Improves system performance and speed.
- ✓ Helps in multitasking.
- ✓ RAM allows smooth running of applications.

Disadvantages of Primary Memory

- ✗ Limited storage capacity.
- ✗ Data is lost in volatile memory (RAM) when power is off.
- ✗ More expensive than secondary memory.



Secondary memory is used to store data and programs permanently. It is non-volatile and has a much larger capacity than primary memory. The CPU cannot access it directly; data is first loaded into primary memory (RAM) for processing.

Types of Secondary Memory



HDD
(Hard Disk Drive)



SSD
(Solid State Drive)



Pen Drive
(Flash Drive)



Memory Card
(SD Card)



CD/DVD
(Optical Disk)



External Drive
(Portable HDD/SSD)

Features of Secondary Memory

- ✓ Non-volatile (data is retained even when power is off).
- ✓ Large storage capacity.
- ✓ Slower than primary memory.
- ✓ Cheaper per GB.
- ✓ CPU cannot access directly (data is loaded into RAM first).

Advantages of Secondary Memory

- ✓ Provides large storage capacity.
- ✓ Stores data permanently.
- ✓ Useful for backup and long-term storage.
- ✓ Less expensive per GB.
- ✗ Allows easy transfer and sharing of data.

Disadvantages of Secondary Memory

- ✗ Slower access speed.
- ✗ Cannot be accessed directly by CPU.
- ✗ Higher physical size (in most cases).

Difference Between Primary and Secondary Memory

	Point	Primary Memory	Secondary Memory
1.	Meaning	Main memory directly used by the CPU.	Storage used to store data and programs permanently.
2.	Speed	Faster.	Slower than primary memory.
3.	CPU Access	CPU can access it directly.	CPU cannot access it directly; data is loaded into primary memory first.
4.	Volatility	RAM is volatile; ROM is non-volatile.	Generally non-volatile.
5.	Capacity	Usually smaller.	Usually much larger.
6.	Cost	More expensive per GB.	Cheaper per GB.
7.	Examples	RAM, ROM, Cache.	HDD, SSD, Pen Drive, Memory Card, CD/DVD.
8.	Use	Stores data/programs currently being processed.	Stores data, files and programs for long-term use.



Easy to Remember :
Primary = Fast + CPU use + Smaller
Secondary = Slow + Long-term storage + Larger

INPUT DEVICES

Introduction to Computer



What are Input Devices ?

- Input devices are hardware components that allow the user to **enter data and instructions** into a computer.
- They help the computer to receive **information** from the outside world.
- The data entered can be text, numbers, **images**, sound, or other types of information.



Key Points:

- Convert human actions into digital form.
- Send data to the CPU.
- Used for inputting text, commands, images, etc.



Types of Input Devices

Keyboard



- Used to enter text, numbers and commands.
- Most commonly used input device.

Mouse



- Used to point, click, select and drag items on the screen.

Scanner



- Converts printed text and images into digital form.
- Used in OCR (Optical Character Recognition).

Microphone



- Used to input audio and voice.
- Helpful in voice recording and speech recognition.

Webcam



- Captures live video and images.
- Used in video calls and online meetings.

Joystick



- Used for gaming and controlling movement.
- Provides directional input.



Input Devices → Convert Human Action → Into Digital Data

OUTPUT DEVICES

Introduction to Computer



What are Output Devices ?

- Output devices are hardware components that **display or produce** the result of data processed by the computer.
- They convert the digital information into **human-readable form** or a physical form.
- The output can be in the form of text, **images**, audio, or video.



Key Points:

- Receive data from the CPU.
- Show the result to the user.
- Can be visual, audio or physical form.



Types of Output Devices

Monitor



- Displays visual output (text, images, video).
- Most common output device.

Printer



- Produces hard copy on paper.
- Types: Inkjet, Laser, Dot Matrix.

Speakers



- Produces sound output.
- Used for music, videos, alerts, etc.

Projector



- Projects visual output on a large screen.
- Used in classrooms, meetings and presentations.

Headphones



- Provides private audio output.
- Used for listening to music or videos.

Plotter



- Produces large, high-quality graphics.
- Used in CAD and engineering drawings.



Output Devices → Convert Digital Data → Into Human Readable Form

INTRODUCTION TO NUMBER SYSTEM

Number system is a method of representing numbers with the help of a set of symbols and rules; it is a mathematical notation used to represent quantities or values in various forms.

TYPES OF NUMBER SYSTEMS

NUMBER SYSTEM

DECIMAL NUMBER SYSTEM

BASE 10

DIGITS
0 – 9

EXAMPLE

$(102)_{10}$, $(24.579)_{10}$, ...

BINARY NUMBER SYSTEM

BASE 2

DIGITS
0 – 1

EXAMPLE

$(11011)_2$, $(101.0011)_2$, ...

OCTAL NUMBER SYSTEM

BASE 8

DIGITS
0 – 7

EXAMPLE

$(245)_8$, $(212.57)_8$, ...

HEXADECIMAL NUMBER SYSTEM

BASE 16

DIGITS
0 – 9, A – F

EXAMPLE

$(2B)_{16}$, $(D3F)_{16}$, ...

IMPORTANT POINTS

- ▶ Humans understand decimal, and computers understand binary, so we have two different number systems.
- ▶ Binary can be long for a large amount of data, so there are more systems like octal and hexadecimal that allow binary sequences to be broken into groups, making them less error-prone and more readable compared to binary.

Introduction to Binary, Octal, Hexadecimal System

1 Decimal Number System

Definition :

- The decimal number system is the standard system for denoting numbers.
- It is also called the base-10 system.
- Digits used in it are 0, 1, 2, 3, 4, 5, 6, 7, 8, 9.
- Each digit in the number is at a specific place value that is powers of 10.
- From right to left – units has the place value as 10^0 , tens has the place value as 10^1 , hundreds as 10^2 , thousands as 10^3 , and so on.

Place Value (Base 10)

Thousands	Hundreds	Tens	Units
10^3	10^2	10^1	10^0
1	0	2	8 5

Example :

10285 can be written as
 $10285 = (1 \times 10^4) + (0 \times 10^3) + (2 \times 10^2) + (8 \times 10^1) + (5 \times 10^0)$
 $= 10000 + 0 + 200 + 80 + 5$
 $= 10285$

2 Binary Number System.

Definition :

- The binary number system is the number system with base 2.
- The numbers are formed using two digits – 0 and 1.
- Binary number system is very useful in electronic devices and computer systems because it can be easily performed using just two states i.e. 0 and 1.
- Each digit in the number is at a specific place value that is powers of 2.
- From right to left – as powers of 2 i.e. 2^0 , 2^1 , 2^2 , etc.

Binary numbers can be converted to Decimal value by multiplying each digit with the place value and then adding the result.

Place Value (Base 2)

2^3	2^2	2^1	2^0
1	0	1	1

Example :

$(1011)_2$ can be written as
 $(1011)_2 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$
 $(1011)_2 = 8 + 0 + 2 + 1$
 $(1011)_2 = 11$ (in decimal)

3 Octal Number System

Definition :

- Octal number system is the number system with base 8.
- The numbers are formed using 8 digits i.e. 0, 1, 2, 3, 4, 5, 6, 7.
- Octal number system is useful for representing file permissions in Unix/Linux operating systems.
- Each digit in the number is at a specific place value that is powers of 8.
- From right to left – as powers of 8 i.e. 8^0 , 8^1 , 8^2 , etc.

Octal numbers can be converted to Decimal value by multiplying each digit with the place value and then adding the result.

Place Value (Base 8)

8^2	8^1	8^0
3	2	5

Example :

$(325)_8$ can be written as
 $(325)_8 = 3 \times 8^2 + 2 \times 8^1 + 5 \times 8^0$
 $(325)_8 = 192 + 16 + 5$
 $(325)_8 = 213$ (in decimal)

4

Hexadecimal Number Systems

Definition :

- Hexadecimal number system is the number system with base 16.
- The numbers are formed using 16 digits i.e. 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, A, B, C, D, E and F.
- Hexadecimal number system is useful for handling memory address locations.
- The digits from 0 to 9 are used as in the decimal system, but the numbers 10 to 15 are represented using the letters A to F as follows.

10 = A, 11 = B, 12 = C, 13 = D, 14 = E, 15 = F



Hexadecimal number system can be converted to Decimal value by multiplying each digit with the place value and then adding the result.

Place Value (Base 16)

16^2	16^1	16^0
2	2	F

Example :

$(2F)_{16}$ can be written as
 $(2F)_{16} = 2 \times 16^1 + F \times 16^0$
 $(2F)_{16} = 2 \times 16 + 15 \times 1$
 $(2F)_{16} = 32 + 15$
 $(2F)_{16} = 47$ (in decimal)



Number Systems - Summary

Number System	Base (Radix)	Digits Used	Place Value (From right to left)	Example	Uses
Decimal	10	0 - 9	$10^0, 10^1, 10^2, 10^3, \dots$	$(10285)_{10}$	General counting in everyday life.
Binary	2	0 - 1	$2^0, 2^1, 2^2, 2^3, \dots$	$(1011)_2$	Used in computers and digital electronics.
Octal	8	0 - 7	$8^0, 8^1, 8^2, 8^3, \dots$	$(325)_8$	Used in UNIX/Linux file permissions.
Hexadecimal	16	0 - 9, A - F	$16^0, 16^1, 16^2, 16^3, \dots$	$(2F)_{16}$	Used in memory addresses, color codes, etc.



Key Takeaways

- The place value of a number system is based on its base (radix).
- Decimal $\rightarrow$ base 10, Binary $\rightarrow$ base 2, Octal $\rightarrow$ base 8, Hexadecimal $\rightarrow$ base 16.
- Higher base systems are useful for storage, memory and processing in computers.
- Each system has a fixed set of digits and follows positional notation.



NUMBER SYSTEM CONVERSIONS

(Decimal ↔ Binary ↔ Octal ↔ Hexadecimal)

Page 1/2

- 1 Decimal → Binary 2 Binary → Decimal 3 Decimal → Octal 4 Octal → Decimal 5 Octal → Binary 6 Binary → Octal 7 Hexadecimal Conversion

1 DECIMAL TO BINARY CONVERSION

Method:

- Divide the decimal number by 2.
- Write the remainder.
- Continue until quotient becomes 0.
- Read the remainders from bottom to top.

Example : Convert 198_{10} to binary

Division	Quotient	Remainder
$198 \div 2$	99	0
$99 \div 2$	49	1
$49 \div 2$	24	1
$24 \div 2$	12	0
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Read
Bottom
to Top

$$198_{10} = 11000110_2$$

Decimal to Binary – 10 Examples

No.	Decimal	Binary	No.	Decimal	Binary
1.	198	11000110_2	6.	625	1001110001_2
2.	13	1101_2	7.	53,625	110101.101_2
3.	231	11100111_2	8.	516.16	1000000100.0010100011_2
4.	932	1110100100_2	9.	156.25	10011100.01_2
5.	85	1010101_2	10.	96	1100000_2

Fractional Part Conversion (Multiply by 2)

$$\begin{aligned} 0.625 \times 2 &= 1.25 \rightarrow 1 \\ 0.25 \times 2 &= 0.50 \rightarrow 0 \\ 0.50 \times 2 &= 1.00 \rightarrow 1 \end{aligned}$$

Therefore,

$$53.625_{10} = 110101.101_2$$

3 DECIMAL TO OCTAL CONVERSION

Method:

- Multiply each decimal number by 8.
- Write the remainder.
- Continue until quotient becomes 0.
- Read bottom to top.

Example : $156_{10} = (?)_8$

$$\begin{aligned} 156 \div 8 &= 19 \text{ R } 4 \\ 19 \div 8 &= 2 \text{ R } 3 \\ 2 \div 8 &= 0 \text{ R } 2 \end{aligned}$$

Read
Bottom
to Top

$$156_{10} = 234_8$$

Decimal to Octal – 7 Examples

No.	Decimal	Octal	No.	Decimal	Octal
1.	156	234_8	6.	175	257_8
2.	25	31_8	7.	198	306_8
3.	64	100_8	8.	255	377_8
4.	100	144_8	9.	512	1000_8
5.	320	500_8	10.	1024	2000_8

7 HEXADECIMAL CONVERSION

Method:

1. Group binary digits in 4s (from right). 2. Convert each group to hex digit.

Example : $(11010110)_2 \rightarrow (?)_{16}$
 $1101 \ 0110 \rightarrow D6_{16}$

$$1101 \ 0110 \rightarrow D6_{16}$$

4 OCTAL TO DECIMAL CONVERSION

Method:

- Multiply each octal digit by its corresponding power of 8 (from right to left starting with 8^0).
- Add all the results.

Example : $(234)_8 = (?)_{10}$

$$\begin{aligned} &= 2 \times 8^2 + 3 \times 8^1 + 4 \times 8^0 \\ &= 2 \times 64 + 3 \times 8 + 4 \\ &= 128 + 24 + 4 \\ &= 156 \end{aligned}$$

$$(234)_8 = 156_{10}$$

Octal to Decimal – 7 Examples

No.	Octal	Decimal	No.	Octal	Decimal
1.	234_8	156_{10}	5.	176_8	126_{10}
2.	31_8	25_{10}	6.	306_8	198_{10}
3.	100_8	64_{10}	7.	377_8	255_{10}
4.	144_8	100_{10}			

Binary ↔ Hexadecimal – 8 Examples

No.	Binary	Hex	No.	Binary	Hex
1.	0000	A_{16}	5.	0100	4_{16}
2.	0001	1_{16}	6.	0101	5_{16}
3.	0010	2_{16}	7.	0110	6_{16}
4.	0011	3_{16}	8.	0111	7_{16}

NUMBER SYSTEM CONVERSIONS

(Decimal ↔ Binary ↔ Octal ↔ Hexadecimal)

Page 2/2

- 1 Decimal → Binary 2 Binary → Decimal 3 Decimal → Octal 4 Octal → Decimal 5 Octal → Binary 6 Binary → Octal 7 Hexadecimal Conversion

2 BINARY TO DECIMAL CONVERSION

Method:

- Multiply each binary digit by its corresponding power of 2 (from right to left starting with 2^0).
- Add all the results.

Example : $(10001)_2 = (?)_{10}$

$$\begin{aligned} &= 1 \times 2^4 + 0 \times 2^3 + 0 \times 2^2 + 0 \times 2^1 + 1 \times 2^0 \\ &= 16 + 0 + 0 + 0 + 1 \\ &= 17 \end{aligned}$$

$$(10001)_2 = 17_{10}$$

Important Powers of 2

2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7	2^8	2^9
1	2	4	8	16	32	64	128	256	512



Binary is
the language
of computers.

BINARY TO DECIMAL – 7 EXAMPLES

No.	Binary	Decimal	No.	Binary	Decimal
1.	10001_2	17_{10}	5.	100110_2	38_{10}
2.	101010_2	42_{10}	6.	11000110_2	198_{10}
3.	10101011_2	171_{10}	7.	110101011_2	845_{10}
4.	1101_2	13_{10}			

5 OCTAL TO BINARY CONVERSION

Method:

- Convert each octal digit into its equivalent 3-bit binary number.

Example : $(573)_8 = (?)_2$

$$5 = 101 \quad 7 = 111 \quad 3 = 011$$

$$(573)_8 = 10111011_2$$

OCTAL TO BINARY TABLE

Octal	Binary (3 bits)	Octal	Binary (3 bits)
0	000	5	100
1	001	6	110
2	010	7	111
3	011		

OCTAL TO BINARY – 7 EXAMPLES

No.	Octal	Binary	No.	Octal	Binary
1.	234_8	1011100_2	5.	175_8	1111101_2
2.	31_8	11001_2	6.	306_8	11000110_2
3.	100_8	100000_2	7.	377_8	1111111_2
4.	144_8	1001000_2			

6 BINARY TO OCTAL CONVERSION

Method:

- Group the binary digits into groups of 3 bits from right. If leftmost group has less than 3 bits, add leading zeros.
- Convert each group to octal equivalent.

Example : $(10011100)_2 = (?)_8$

- Step 1 : Group into 3 bits from right
 $100 \ 111 \ 00$
 Step 2 : Add leading zeros (if needed)
 $010 \ 011 \ 100$
 Step 3 : Convert each group to octal
 $4 \ 3 \ 4$

$$(10011100)_2 = (434)_8$$

BINARY TO OCTAL – 7 EXAMPLES

No.	Binary	Octal	No.	Binary	Octal
1.	10011100_2	234_8	5.	1111101_2	175_8
2.	11001_2	31_8	6.	11000110_2	306_8
3.	100000_2	100_8	7.	1111111_2	377_8
4.	1100100_2	144_8			

6 HEXADECIMAL CONVERSION (ADDITIONAL)

Binary → Hexadecimal

- Group 4 bits (nibbles) from right, convert to hex digit.

Example : $(110111001010)_2$
 $1101 \ 1100 \ 1010$
 $\rightarrow D \ C \ A$

Decimal → Hexadecimal

- Method: Divide by 16 and read remainders from bottom to top.

Example : $(255)_{10}$
 $255 \div 16 = 15 \text{ R } 15 \text{ (F)}$
 $15 \div 16 = 0 \text{ R } 15 \text{ (F)}$
 $(255)_{10} = (FF)_{16}$

Hexadecimal → Decimal

- Method: Multiply each hex digit by its power of 16 and add results.

Example : $(3A)_{16}$
 $3 \times 16^1 + 10 \times 16^0$
 $= 48 + 10 = 58$
 $(3A)_{16} = 58_{10}$

Hexadecimal Digits

0	1	2	3	4	5	6	7
8	9	A	B	C	D	E	F

Quick Reference – Conversion Rules

- Decimal ↔ Binary ($\div 2 / \times 2$)
 Decimal ↔ Octal ($\div 8 / \times 8$)
 Decimal ↔ Hex ($\div 16 / \times 16$)
 Binary ↔ Octal (group 3 bits)
 Binary ↔ Hex (group 4 bits)
 Octal ↔ Binary (3 bits)

BINARY ARITHMETIC OR CONVERSION

Simple Addition, Subtraction, Multiplication

1 ADDITION



Definition :

Binary addition is performed in the same manner as decimal addition. However, since binary number system has only two digits, the addition table for binary arithmetic is very simple, consisting of only four entries.

The complete table for binary addition is:

A	B	SUM	CARRY
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Rules :

- $0 + 0 = 0$
- $0 + 1 = 1$
- $1 + 0 = 1$
- $1 + 1 = 0$, plus a carry of 1 to next higher column.

Examples :

1) $(1011)_2 + (1101)_2$

$$\begin{array}{r} 1\ 0\ 1\ 1 \\ +\ 1\ 1\ 0\ 1 \\ \hline 1\ 0\ 0\ 0 \end{array}$$

$(1011)_2 + (1101)_2 = (11000)_2$

2) $(11101)_2 + (10111)_2$

$$\begin{array}{r} 1\ 1\ 1\ 0\ 1 \\ +\ 1\ 0\ 1\ 1\ 1 \\ \hline 1\ 0\ 1\ 1\ 0 \end{array}$$

$(11101)_2 + (10111)_2 = (101100)_2$

3) $(10011)_2 + (11100)_2$

$$\begin{array}{r} 1\ 0\ 0\ 1\ 1 \\ +\ 1\ 1\ 1\ 0\ 0 \\ \hline 0\ 0\ 0\ 1\ 1\ 1 \end{array}$$

$(10011)_2 + (11100)_2 = (10000111)_2$

2 SUBTRACTION



Definition :

Binary subtraction is the process of finding the difference between two binary numbers using the base-2 number system. It follows rules similar to decimal subtraction but limited to digits 0 and 1. It borrows 1 from the next higher order bit to the left.

A	B	A - B	BORROW
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

Rules :

- $1 - 0 = 1$
- $1 - 1 = 0$
- $0 - 0 = 0$
- $0 - 1 = 1$, with a borrow

Examples :

1) $(10110)_2 - (01001)_2$

$$\begin{array}{r} 1\ 0\ 1\ 1\ 0 \\ -\ 0\ 1\ 0\ 0\ 1 \\ \hline 0\ 1\ 0\ 0\ 1 \end{array}$$

$(10110)_2 - (01001)_2 = (01001)_2$

2) $(11001)_2 - (1011)_2$

$$\begin{array}{r} 1\ 1\ 0\ 0\ 1 \\ -\ 1\ 0\ 1\ 1 \\ \hline 1\ 0\ 0\ 1\ 0 \end{array}$$

$(11001)_2 - (1011)_2 = (10010)_2$

3) $(100000)_2 - (1111)_2$

$$\begin{array}{r} 0\ 0\ 0\ 0\ 0\ 0 \\ -\ 0\ 0\ 1\ 1\ 1 \\ \hline 0\ 0\ 1\ 0\ 0\ 1 \end{array}$$

$(100000)_2 - (1111)_2 = (011001)_2$

BINARY ARITHMETIC OR CONVERSION

Simple Addition, Subtraction, Multiplication

3 MULTIPLICATION



Definition :

Binary multiplication is the process of finding the product of two binary numbers using the base-2 number system. It follows the same principle as decimal multiplication but uses only the digits 0 and 1.

Basic Rules of Binary Multiplication

A	B	A × B
0	0	0
0	1	0
1	0	0
1	1	1

Basic Products :

- $0 \times 0 = 0$
- $0 \times 1 = 0$
- $1 \times 0 = 0$
- $1 \times 1 = 1$

Example :

$(2F)_{16}$ can be written as
 $(2F)_{16} = 2 \times 16^1 + F \times 16^0$
 $(2F)_{16} = 2 \times 16 + 15 \times 1$
 $(2F)_{16} = 32 + 15$
 $(2F)_{16} = 47$ (in decimal)

1) $11_2 \times 11_2$ ($3 \times 3 = 9$ in decimal)

$$\begin{array}{r} 1\ 1 \\ \times\ 1\ 1 \\ \hline 1\ 1 \\ 1\ 1\ 0 \\ \hline 1\ 0\ 0\ 1 \end{array}$$

$11_2 \times 11_2 = 1001_2$

2) $1011_2 \times 101_2$ ($11 \times 5 = 55$ in decimal)

$$\begin{array}{r} 1\ 0\ 1\ 1 \\ \times\ 1\ 0\ 1 \\ \hline 1\ 0\ 1\ 1 \\ 0\ 0\ 0\ 0 \\ 1\ 0\ 1\ 1\ 0 \\ \hline 1\ 0\ 1\ 1\ 1\ 1 \end{array}$$

$1011_2 \times 101_2 = 110111_2$

3) $101_2 \times 101_2$ ($5 \times 5 = 25$ in decimal)

$$\begin{array}{r} 1\ 0\ 1 \\ \times\ 1\ 0\ 1 \\ \hline 1\ 0\ 1 \\ 0\ 0\ 0 \\ 1\ 0\ 1\ 0 \\ \hline 1\ 0\ 0\ 1\ 0\ 1 \end{array}$$

$101_2 \times 101_2 = 100101_2$



Quick Revision

Binary Multiplication Rules :

- $0 \times 0 = 0$
- $0 \times 1 = 0$
- $1 \times 0 = 0$
- $1 \times 1 = 1$

- ★ Use the same method as decimal multiplication.
- ★ Multiply each digit of the multiplier with the number.
- ★ Shift the partial product according to the place value.
- ★ Add all partial products to get the final result.